

Experiment:- Study of LAN Topology, Network devices and Address configuration.

Objective:-

- 1> To study the LAN topology used in collage campus.
- 2> To identify networking device & understand their function
- 3> To analyze wired & wireless communication method used in the network.
- 4> To measure speed of network for wired & wireless connection.
- 5> To obtained MAC, IP Address & subnet mask.

Problem Statement - 1

Draw and identify LAN topology of collage campus & clearly label all components such as routes, switches & etc.

Solⁿ The college campus network observed this study consist of.

- * Layer-3 core switch in the server room
- * Managed by poe + switches
- * Ethernet port in classroom & Laboratories.

Network topology - Hybrid hierarchical topology which combines

- i> Tree topology.
- ii> Extended star topology.
- iii> Ring topology (server room)

The network hierarchical layer:-

- 1) core layer.
- 2) Distribution layer.
- 3) Access layer.

* core layer:- core layer contain server room and server room have firewall router, 2-L3+switches, patch rack, server, inverter etc.

* Distribution layer:-

→ Distribution layer contains 4 switches.

↳ The four switches are.

- 1) Main Building - R7 (Room no 213, 2nd floor)
- 2) Main Building - R1 (Room no 301, 3rd floor)
- 3) Main Building - R5 (Room no 113, 1st floor)
- 4) Main Building - R2 (Room no 201, 2nd floor)

↳ This four switches contains.

1) 18 port gigabit easy smart switch with 16-port
TL-SG1218MPE

② SG2418XMP - 24 port gigabit & 4 port
10Gbps, L2+Managed switches with 24 port
PoE+.

↳ TP-Link TL-SG1218MPE:-

- * Likely used for: CCTV camera, wifi access point etc
- * 18 port gigabit "Easy smart" switch
- * 16 port support PoE

② TP-Link Omada SG3428XMP

* Higher-end managed layer 2+switch.

* 24 PoE+ port, uplinks, 10Gbps

* Usually act as a main distribution for each floor.

* Access Layer

↳ Access Layer contains 3-switches i.e.

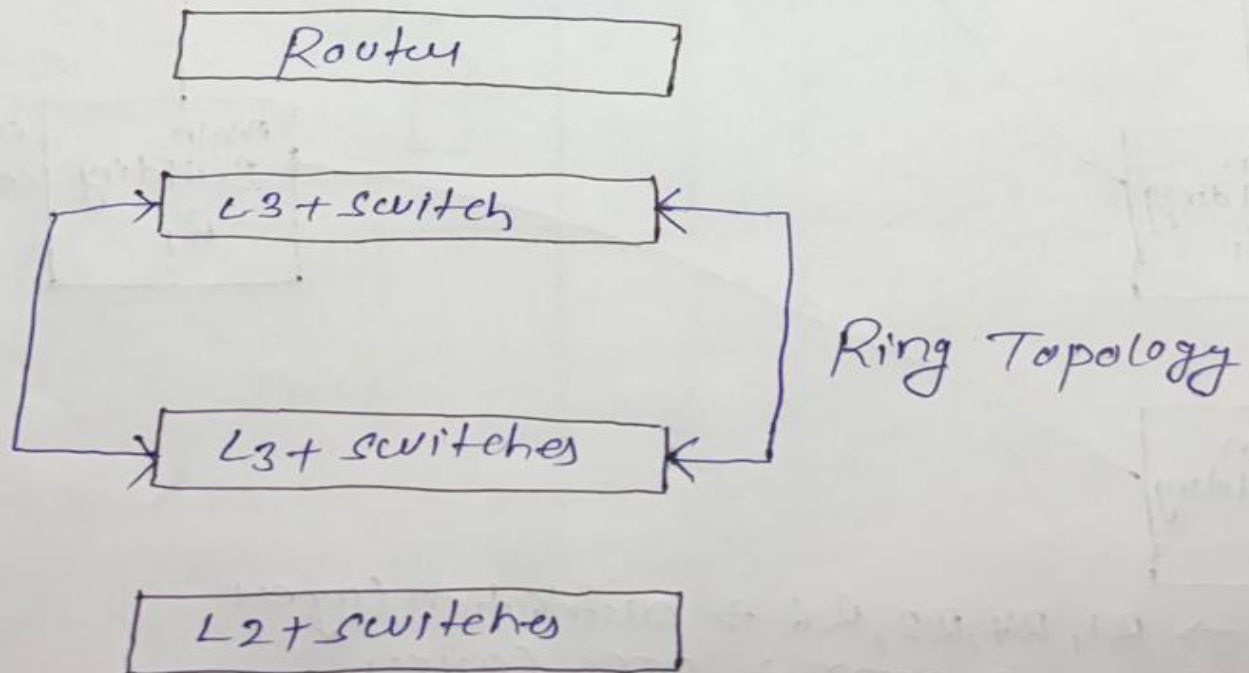
- 1) Main Building R₁ - [Room no - 301, 3rd floor]
- 2) Main Building R₄ - [Room no - 05, ground floor]
- 3) Main Building R₅ - [Room no 18, ground floor]

↳ Every switch act as access layer which helps to provide network connectivity to all classroom lab faculty room, etc.

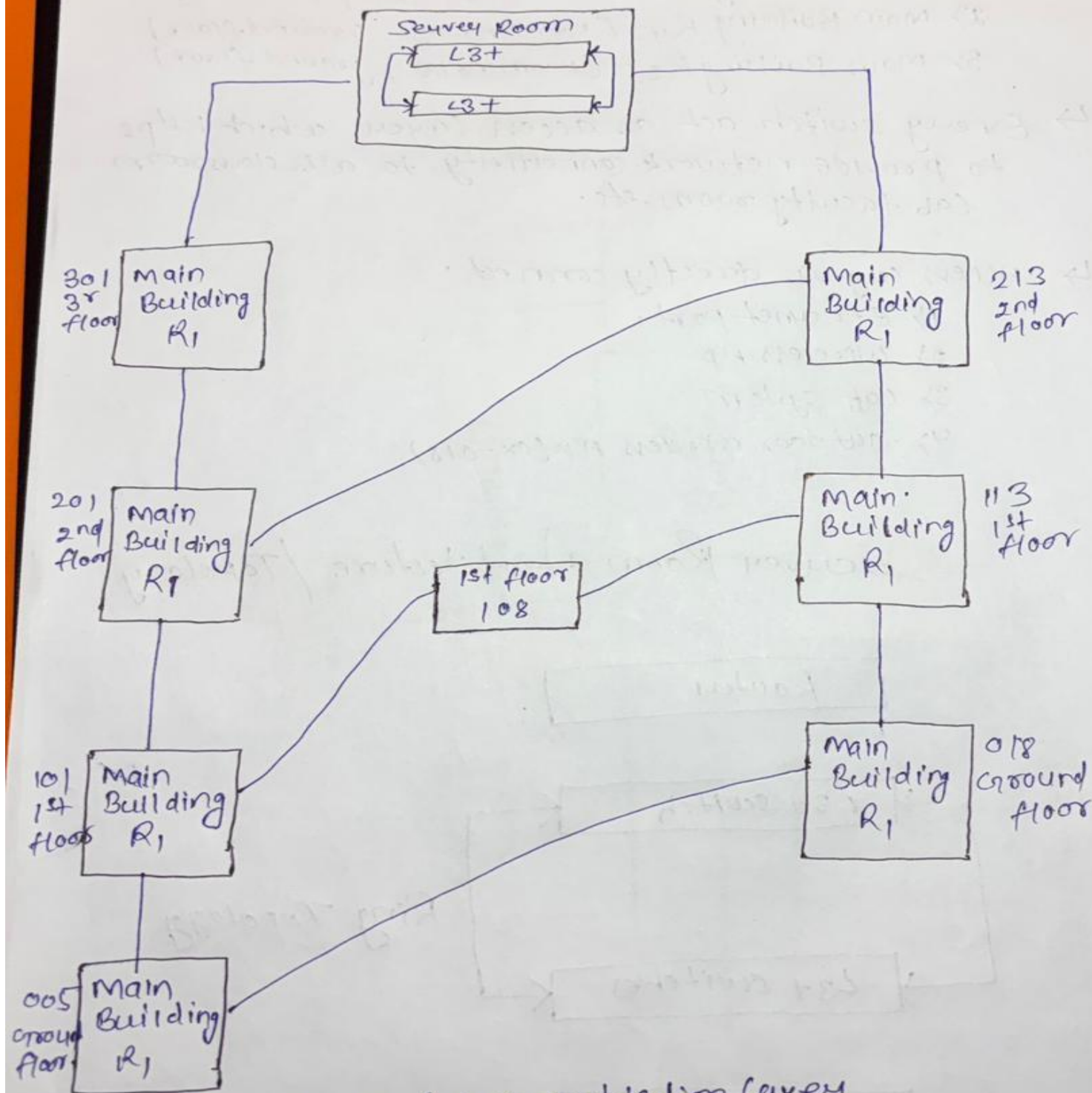
↳ Access layer directly connect.

- 1) Ethernet port.
- 2) Wireless Ap
- 3) Lab system
- 4) Outdoor wireless Ap (ex-018)

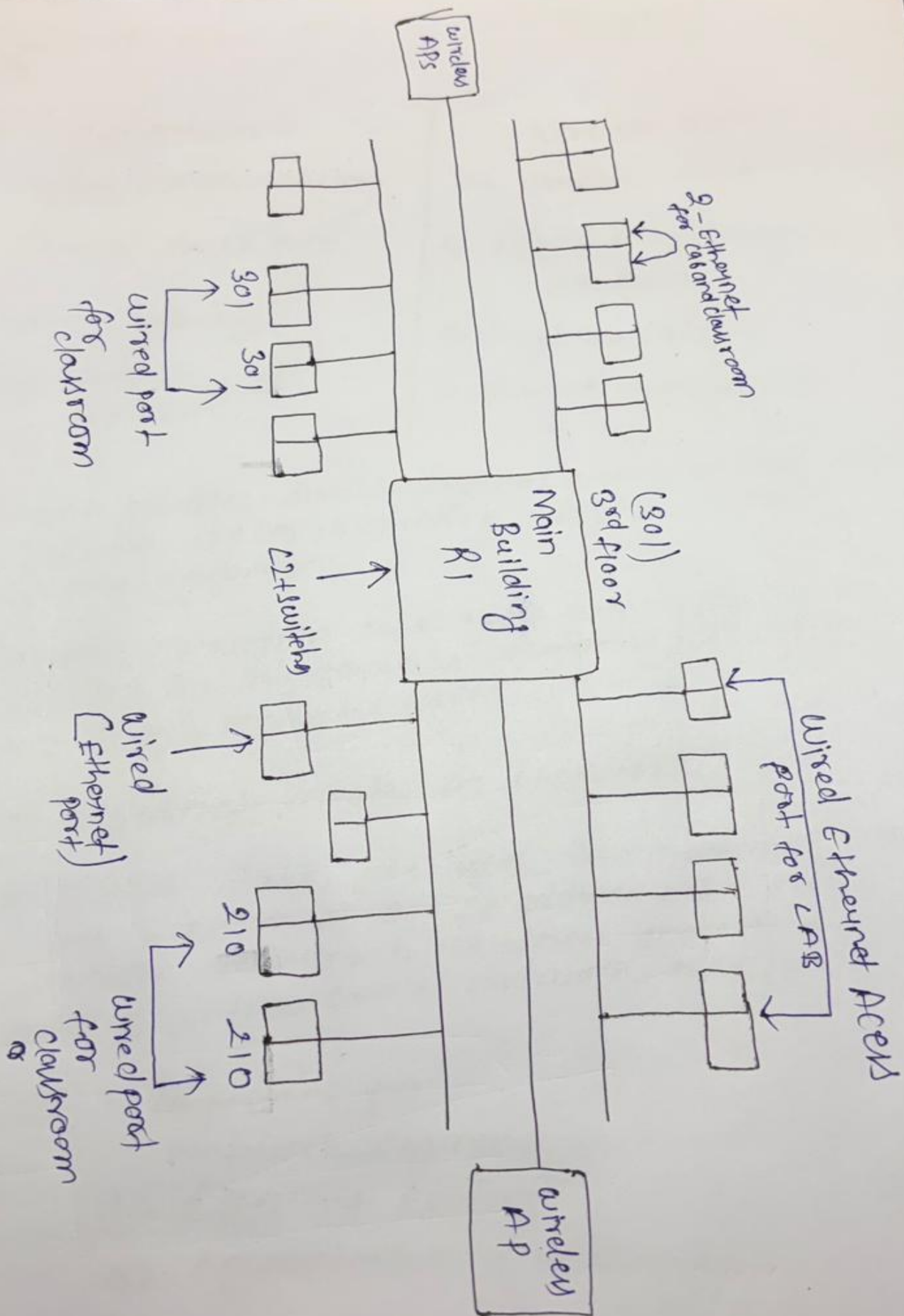
Server Room Architecture / Topology



Topology



- R1, R7, R2, R6 → Distribution Layer
- Server room → Core Layer
- R1, R2, R3, R4, R5, R6, R7 → Access Layer.



Question 3: Analyze Connection Types:

- Specify whether connections are:
 - Wired (Ethernet)
 - Wireless (Wi-Fi)
- Mention where each type is used.

```
PS C:\Users\kamar> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 7:

    Connection-specific DNS Suffix  . :
    Link-Local IPv6 Address . . . . . : fe80::27b1:44cb:2ace:8df9%17
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:1ca8:7d93:76ed:a8dd:d18b:ad59
    Temporary IPv6 Address. . . . . : 2401:4900:1ca8:7d93:ed48:3707:8ceb:3f92
    Link-Local IPv6 Address . . . . . : fe80::f025:1618:de47:3938%16
    IPv4 Address. . . . . : 192.168.1.110
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::fe9f:2aff:fe3a:4d59%16
                                192.168.1.1

Ethernet adapter Bluetooth Network Connection:

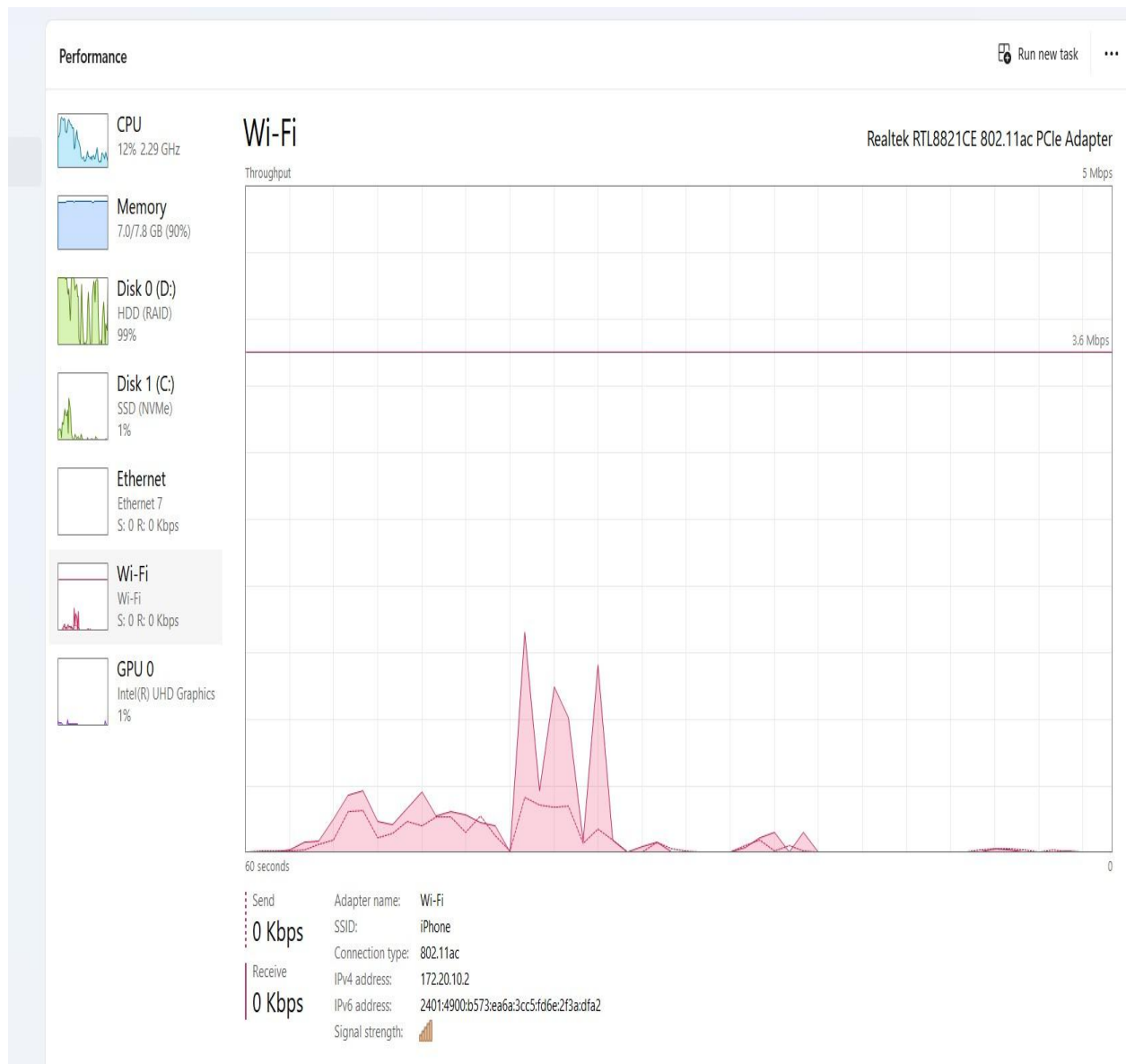
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
PS C:\Users\kamar> |
```

➔ By using ipconfig

QUESTION 4:-

Measure Network Speed:

- Find the link speed of:
 - Wired network (if available)
 - Wireless network
- Mention the tools/commands used (e.g., Task Manager, netsh; netsh wlan show interfaces or , etc.).



- Mention the tools/commands used (e.g., Task Manager, netsh; netsh wlan show interfaces or , etc.).

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\kamar> netsh wlan show interfaces

There is 1 interface on the system:

    Name                           : Wi-Fi
    Description                     : Realtek RTL8821CE 802.11ac PCIe Adapter
    GUID                           : 9e393a06-764a-44a8-9dc9-02bc346a3023
    Physical address                : 00:e9:3a:b2:1a:fd
    Interface type                  : Primary
    State                           : connected
    SSID                           : Airtel_Shanvi Library 168
    AP BSSID                       : fc:9f:2a:3a:4d:60
    Band                           : 5 GHz
    Channel                        : 48
    Connected Akm-cipher           : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
    Network type                   : Infrastructure
    Radio type                     : 802.11ac
    Authentication                  : WPA2-Personal
    Cipher                         : CCMP
    Connection mode                 : Auto Connect
    Receive rate (Mbps)            : 433.3
    Transmit rate (Mbps)          : 433.3
    Signal                         : 100%
    Rssi                           : -40
    Profile                       : Airtel_Shanvi Library 168
    QoS MSCS Configured             : 0
    QoS Map Configured              : 0
    QoS Map Allowed by Policy       : 0

PS C:\Users\kamar> |
```

->By using netsh wlan show interfaces

QUESTION 5. Find System Network Details:

- Determine and record the following of your computer:
 - MAC Address
 - IP Address (IPv4)
 - Subnet Mask
- Mention the command(s) used (e.g., ipconfig, getmac).

```
PS C:\Users\kamar> getmac

Physical Address      Transport Name
=====
00-FF-A6-C5-3D-3D    Media disconnected
00-E9-3A-B2-1A-FC    Media disconnected
48-9E-BD-75-FD-03    Media disconnected
00-E9-3A-B2-1A-FD    \Device\Tcpip_{9E393A06-764A-44A8-9DC9-02BC346A3023}
0A-00-27-00-00-11    \Device\Tcpip_{A3A583F6-4438-4A27-97EA-C42D3F6A057F}
PS C:\Users\kamar> |
```

->By using getmac

Difference b/w wired & wireless speed

Soln

Wired Network

- ↳ Faster communication
- ↳ Stable connection
- ↳ Low latency
- ↳ Dedicated physical medium

wireless Network

- ↳ slower communication
- ↳ Affected by distance / interface
- ↳ Higher latency
- ↳ shared radio medium

↳ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

↳ wireless connection use radio waves which are affected by interference, distance obstacles & multiple connected users

Q Role of subnet mask in network?

Soln

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belong to the same network or different

It is important for

- 1) Network segment
- 2) Efficient Routing
- 3) communication b/w devices

→ Observation

During the study of the college network infrastructure

1) A centralised server room was used as network backbone

11) Two L3 switches were connected in redundant cyclic structure.

3) managed T1-LINK PoE+ switches were used in network cabinets

→ Advantage of Existing Network?

* Scalable architecture

* Centralized management.

* Supports PoE+ devices.

* Easy expansion of floors & classroom

* Supports both wireless & wired

* Redundant link for failover.